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Addition of low concentrations of an ionic liquid to a base oil reduces friction over multiple length scales: a combined nano- and macrotribology investigation†

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The efficacy of ionic liquids (ILs) as lubricant additives to a model base oil has been probed at the nanoscale and macroscale as a function of IL concentration using the same materials. Silica surfaces lubricated with mixtures of the IL trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate and hexadecane are probed using atomic force microscopy (AFM) (nanoscale) and ball-on-disc tribometer (macroscale). At both length scales the pure IL is a much more effective lubricant than hexadecane. At the nanoscale, 2.0 mol% IL (and above) in hexadecane lubricates the silica as well as the pure IL due to the formation of a robust IL boundary layer that separates the sliding surfaces. At the macroscale the lubrication is highly load dependent; at low loads all the mixtures lubricate as effectively as the pure IL, whereas at higher loads rather high concentrations are required to provide IL like lubrication. Wear is also pronounced at high loads, for all cases except the pure IL, and a tribofilm is formed. Together, the nano- and macroscales results reveal that the IL is an effective lubricant additive – it reduces friction – in both the boundary regime at the nanoscale and mixed regime at the macroscale.

Introduction

Tribological effects, such as friction and wear, reduce the energy efficiency and working life of machinery, and have been estimated to cost 1.5% of GDP in industrialised countries.[1] Lubricants play an important role in minimising friction and wear by acting as a barrier between moving surfaces. Conventional lubricants are categorised as natural (*e.g.* mineral oils) or synthetic (*e.g.* perflouropolyethers, PFPEs).

The mechanisms of lubrication can be divided into four different regimes: hydrodynamic, elastohydrodynamic, mixed and boundary lubrication.[1,2] Which regime operates depends mainly on the distance between the surfaces. At very low contact pressures and/or high speeds, lubrication is in the elastohydrodynamic/hydrodynamic regimes, where the lubricants form a fluid film barrier between the moving surfaces to reduce friction and wear. The friction coefficient (the frictional : normal force ratio, μ) increases with the lubricant viscosity in these regimes.[3] At medium contact pressures and/or speeds, lubrication is in the mixed regime, and there is partial contact of the sliding surfaces. At high contact pressures and/or low speeds, lubrication is in the boundary regime. The sliding surfaces are close together and conventional lubricants can sometimes be squeezed out, resulting in surface contact. As such, in the boundary regime the integrity of the adsorbed layer under pressure determines lubrication performance. Ideally, a molecularly thin layer of the lubricant (or lubricant additive) is strongly adsorbed to the sliding surfaces to support the load which prevents wear.[3]

In modern machines, such as internal combustion engines, lubrication is mainly in the hydrodynamic and elastohydrodynamic regimes, where the main cause of friction is hydrodynamic drag. This consumes 10–15% of the total engine generated energy and is proportional to the lubricant viscosity.[4] One method to increase engine efficiency is to use less viscous lubricants. Lower viscosity moves the system into the mixed or boundary regimes. In these regimes the surfaces are less protected because conventional lubricants only weakly adhere to the solid surfaces. This can lead to surface contact and significant wear. Additives, usually containing long, polar functional groups, are commonly mixed with conventional lubricants (base oils) to form protective films

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and improve lubrication performance. Zinc dialkyldithiophosphate (ZDDP) is a commonly used additive, but it generates deposits from thermal decomposition and is mainly used for steel surfaces where it forms a tribofilm.[5]

Ionic liquids (ILs) are salts composed entirely of ions which are liquid at temperatures less than 100 °C.[6–8] They are attractive substances because their properties can be predictably tuned for a given application through systematic changes in ion structure.[9,10] ILs are high performing lubricants in neat form,[3,11–14] and potentially excellent lubricant additives,[5,15,16] especially in the boundary regime, due to their strong affinity to solid surfaces.[17–24] IL ions interact with surfaces and each other *via* Coulombic, van der Waals, H-bonding and solvophobic interactions[25,26] which means they form robust boundary layers and resist 'squeeze-out'.[27] This means IL boundary layers remain in place up to higher loads than those of comparable conventional lubricants.[20,28–30] The generally high mechanical and thermal stability of ILs means they resist decomposition, and high thermal conductivity means ILs transfer heat away from the sliding surfaces more effectively than conventional lubricants.[31]

Compared with current commercially available lubricants, like mineral oils, ILs are expensive, so the cost benefit ratio is unfavourable, except for high end applications. However, the recent development of oil-miscible ILs means they can be used as lubricant additives.[16,32] ILs have distinct advantages over traditional lubricant additives such as ZDDP due to their stability at high temperatures, ashless decomposition, and compatibility with a wider range of surfaces, including steel, aluminium, titanium and silica.[5,33–35]

Macrotribology studies have shown IL–oil mixtures reduce friction and wear more effectively than neat mineral or synthetic base oils in several systems.[4,5,16,32,33,35–38] However, the mechanism by which IL additives lubricate a surface remains only partially understood.[33] Nanotribology studies can assist in the elucidation of lubrication mechanisms because the influences of surface asperities and roughness are reduced.

We have recently shown that pure ILs are much more effective lubricants than base oils for sliding silica surfaces at the nanoscale.[15] More importantly, as little as 1.0 mol% IL dissolved in base oil lubricated as effectively as the pure IL, and similar results have been obtained at the macroscale for different ILs and surfaces.[33,37] However, for IL–oil mixtures, links between macrotribolgy and nanotribology have not been established, partially because the same system has not been studied at different length scales. In this work, we address this issue by correlating nanoscale and macroscale tribology results for silica surfaces lubricated with mixtures of trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate ($P_{6,6,6,14}(^iC_8)_2PO_2$) and hexadecane under similar initial Hertzian contact stresses.

Materials and methods

Silicon wafers were purchased from Silicon Valley Microelectronics CA, USA. The surface roughness is 0.5 ± 0.3 nm. Hexadecane with a purity of ReagentPlus®, 99% was purchased from Sigma Aldrich. The IL trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)-phosphinate (CAS 465527-59-7, molecular weight 773.27 g mol^{-1}, purity > 95%) was supplied by Iolitec. The viscosity of this IL at 25 °C is ~1000 mPa s, the melting point is −71 °C, and the thermal decomposition temperature is 135 °C.[39]

Nanotribological measurements were performed using a Digital Instruments NanoScope VIII Multimode AFM with an EV scanner in contact mode on silicon wafers. Two sharp silicon tips (spring constant = 2.0 ± 0.2 N m^{-1}, tip radius ~8 nm) from the same batch (model NSC36, Mikromasch, Tallinn, Estonia) were used over the course of the investigation. The tip was cleaned immediately prior to use by careful rinsing in Milli-Q water, drying under nitrogen and irradiation with ultraviolet light for 20 min. 0.1 mL of the sample was injected in the liquid cell every time. Lateral force measurements were acquired in contact mode by performing lateral AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled.[40] The scan size was 100 nm, and scan rate was 30 Hz. The lateral deflection signal (*i.e.*, cantilever twist) was converted to lateral force using a customized function produced in Matlab 7.0 which takes into account the torsional spring constant and the geometrical dimensions of the cantilever.[29] These experiments were repeated on more than five occasions, and data with the same form but with small variations in the lateral force was obtained. The water content of the IL/hexadecane mixtures were measured using Karl–Fischer titration before and after AFM experiments. They were below 0.03 wt% before the experiment and below 0.06 wt% after the experiment.

The macrotribology tests were conducted at room temperature on a Nanovea (Irvine, CA) ball-on-disk tribometer using 3 mm radius Si_3N_4 balls on silicon wafers with a sliding speed of 50 mm s^{-1}, and sliding distance of 20 m. The surface roughness of the Si_3N_4 is 2 ± 1 nm. The friction coefficient was recorded throughout the experiment. Experiments were conducted at loads of 2 N and 10 N. A series of IL–hexadecane mixtures were added to the silicon wafer before the loaded Si_3N_4 ball was placed on its surface. On completion of wear tests the wear scar was measured using a NeoScope JCM-5000 scanning electron microscope; the wear depth was measured using a Bruker GT-K1 Optical Profiler.

Results and discussion

$P_{6,6,6,14}(^iC_8)_2PO_2$ is chemically stable to both air and water and fully miscible with apolar base oils.[41] Hexadecane is used as a model apolar base oil because of its simple structure and well-known properties, which aids interpretation of results. IL–hexadecane mixtures with IL concentrations between 0 to 10 mol% were investigated using atomic force microscopy (AFM) (nanotribology) and ball-on-disc tribometer (macrotribology) measurements.

Hertzian contact stress refers to the stress close to the area of contact between two surfaces of different radii. In this work the Hertzian contact stresses are comparable which facilitates comparison of nanotribology to macrotribology results.

Calculations of the Hertzian contact stresses for both length scales are shown in the ESI.†

Nanotribology

The nanotribology of pure hexadecane, pure IL and the IL–hexadecane mixtures was investigated using AFM. The lateral (frictional) force (F_L) as a function of normal load (F_N) was obtained using sharp silicon AFM tips ($r \approx 8$ nm) on silica, *cf.* Fig. 1. The lateral forces are highest for hexadecane, which is approximately half that measured in air,[15] and the pure IL reduces friction by approximately 50% compared to pure hexadecane. Data for the different IL–hexadecane mixtures falls between that of pure hexadecane and pure IL.

The friction coefficient (μ) is obtained from a modified version of Amontons' law, $F_L = F_L(0) + \mu F_N$, where $F_L(0)$ is the lateral force at zero applied normal load, and mostly depends on the lubricant, and the roughness and compressibility of the AFM tip and the surface.[42,43] μ is extracted from the gradient of the lateral force-normal load data in the linear regime and thus is not dependent on adhesion. The friction coefficients for the pure liquids and mixtures in the linear regime are reported in Table 1. The friction coefficient in hexadecane (μ = 0.57) is lower than that in air, (μ = 1.0) and in octane (μ = 1.1) determined previously.[15] The similarity of the friction coefficients in air and octane was attributed to the tip surface contact in both systems, *i.e.* the octane is displaced even at low loads. The lower friction for hexadecane means that it is not displaced as readily as octane, which may be attributed to its higher viscosity.

For the IL–hexadecane mixtures, adhesion decreases with IL concentration, as does the lateral force. A measure of the adhesion is manifested as the intercept of the lateral force data with the abscissa. The adhesion can be considered as an additional load acting between the surfaces; this effect is generally only observed at the nanoscale.[44] For both the pure liquids and the mixtures, the rate of increase in lateral force with normal load in the linear region (above ~30 nN) is constant except for the lower concentrations of 0.01 mol% IL and 1.0 mol% IL in hexadecane mixtures. For these concentrations, when the normal load increases above ~180 nN and ~200 nN (respectively), the lateral force increases more rapidly. For IL concentrations 2.0 mol% and above lateral forces are comparable with the pure IL. These results are similar to those seen for IL–octane mixtures and are consistent with an IL boundary layer adsorbed to the silica surface at lower concentrations that may be displaced at higher loads.

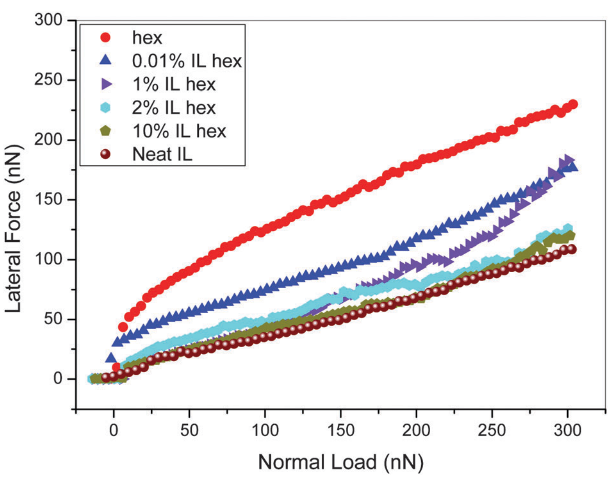

Fig. 1 Lateral force *versus* normal load in IL–hexadecane mixtures for a sharp silicon AFM tip (r = 8 nm) sliding on a silicon wafer at 6 μm s^{-1} up to 300 nN.

Table 1 Nanoscale friction coefficients of IL–hexadecane mixtures in the linear regime

	Hexadecane	0.01% IL	1% IL	2% IL	10% IL	Pure IL
μ	0.57	0.40	0.40	0.37	0.39	0.34

For IL–oil mixtures, it has previously been demonstrated that the IL surface excess increases with the bulk IL concentration.[15] For low IL concentrations, the surface excess is low, and the ions in the boundary layer are relatively far apart which leads to a fragile boundary layer that is ruptured at low forces. It has been shown earlier that the normal force required to remove the boundary layer from a contact depends on the surface excess[15,45] and the degree of shear[46] At higher bulk IL concentrations the surface is saturated with IL, meaning the boundary layer ions are tightly packed and thus their resistance to collective displacement increases. The result is a robust boundary layer that remains in place up to large normal force, as observed here for IL concentrations of 2.0 mol% and above. For 0.01 mol% and 1.0 mol% IL, the boundary layer remains in place at low loads, but is disrupted at higher loads resulting in friction increasing towards that of the pure hexane case. The fact that the increase in friction is gentle rather than step-like reveals that disruption of the boundary layer occurs gradually and implies a certain self-healing after collapse, as catastrophic collapse of a boundary film causes a clear step change in friction.[47] The boundary layer is thus apparently replenished on a time scale of the same order as the experiment when it is not directly confined by the AFM tip. As the normal force on the tip is increased, more ions are displaced from the boundary layer during sliding, and recovery is incomplete over the time it takes for the tip to depart and return, indicating that ion adsorption is relatively slow in the viscous base oil.

Macrotribology

The macrotribological performance of pure hexadecane, pure IL and IL–hexadecane mixtures on silicon wafers were tested using a pin-on-disk tribometer with a sliding speed of 50 mm s^{-1} and loads of 2 N and 10 N. The 2 N load corresponds to a Hertzian contact stress of 1.2 GPa, which is approximately equal to a load of 160 nN in the nanotribology data in Fig. 1. A 10 N load corresponds to a Hertzian contact stress of 2.1 GPa and is thus comparable with the 300 nN load at nanoscale, assuming that the tip dimensions are not significantly altered during the nanotribology experiments and that wear of the ball used in the macrotribology experiments is low. Given the difficulties of monitoring these parameters during the course of the experiment, the initial contact pressures were matched. The macroscale friction coefficients *versus* sliding distance for IL–hexadecane mixtures are shown in Fig. 2. Comparison of the nanoscale friction coefficients in Table 1 with those in Fig. 2 and Table 2 reveals that macroscale friction coefficients are about 5 times smaller. As the surfaces, the lubricants and the pressures are initially comparable at the two scales, the results reveal that the mechanisms and regimes are intrinsically different for the multi-asperity contact at macroscale and the single-asperity contact at the nanoscale.

The thickness of the lubricant film is a key factor controlling which lubrication regime operates. The film thicknesses for the nanoscale AFM and macroscale ball-on-disk experiments employed here were calculated using the Hamrock and Dowson model,[1,48] (*cf.* Table S1, ESI†). The details of this model are provided in the ESI.† The pressure-viscosity coefficient is a property that defines the capacity of the lubricants, and is independent of the contact surfaces.[49] The pressure-viscosity coefficient used in these calculations is unknown so assumptions have been made,[50–53] which are detailed in the ESI.† The effect of varying this parameter over a range of reasonable values is also shown, and while the thickness values obtained vary somewhat, the conclusions do not. λ is the ratio of the expected lubricant thickness (shown in ESI,† Tables S1 and S2) and the composite surface roughness. λ is considered to be less than 1.2 for boundary lubrication, between 1.2 and 3 for mixed lubrication, between 3 and 10 for elastohydrodynamic lubrication, and above 10 for hydrodynamic lubrication.[2] Because the same substrate is employed at both length scales and the roughness of the AFM tip and macrotribology ball are comparable, a tribopair composite surface roughness of $\sim$2.0 nm is used for both length scales. At the macroscale, λ is calculated to be between 1.2–1.8 for hexadecane and IL–hexadecane mixtures, and higher than 60 for the pure IL. These values did change appreciably when the load was increased from 2 nN to 10 nN. The higher λ for the pure IL is a consequence of the large expected film thickness which in turn is a result of its much higher viscosity (see eqn (S4) in ESI†). On the basis of these calculations all of the IL–hexadecane mixtures are in the mixed regime (where both boundary lubrication and elastohydrodynamic effects operate simultaneously), whereas the pure IL is in the hydrodynamic regime. The fact that the friction coefficients are generally higher at 10 N than at 2 N is a consequence of the fact that, according to the generalised Martens–Stribeck curve, the ordinate is inversely proportional to the pressure. Since there is a negative gradient in the mixed lubrication regime, a higher load leads to a higher friction coefficient. (The negative gradient is a consequence of hydrodynamic lift reducing asperity contacts and thus friction; in this regime the increased load increases the number of asperity contacts.)

Table 2 Macroscale friction coefficients of IL–hexadecane mixtures

		Hexadecane	0.01% IL	1% IL	10% IL	Pure IL
μ	2 N	0.11	0.06	0.06	0.05	0.06
	10 N	0.12	0.12	0.12	0.08/0.14	0.08

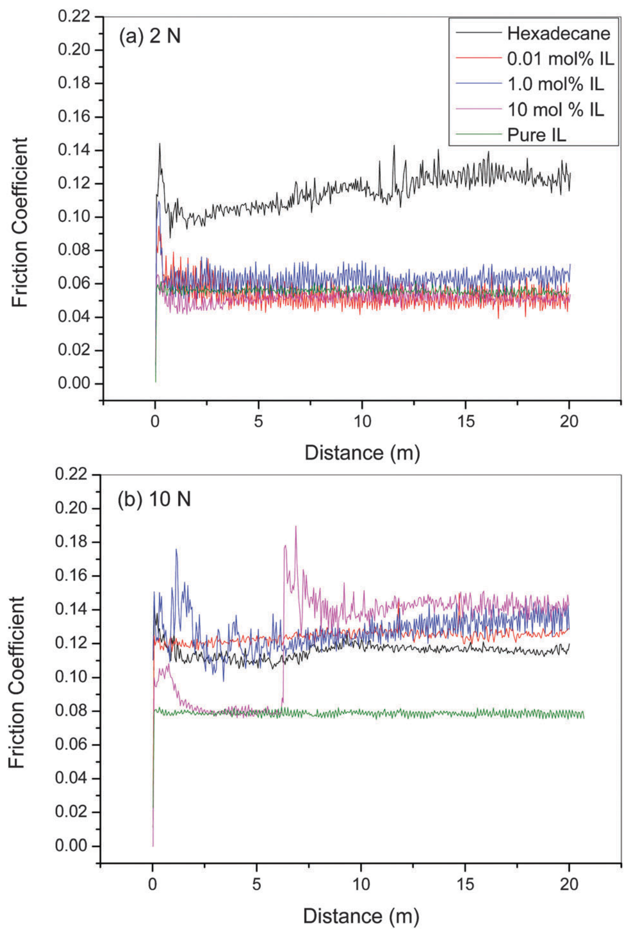

Fig. 2 Macroscale friction coefficient *vs.* sliding distance for a silicon nitride ball sliding on a silicon wafer with a normal load of (a) 2 N, and (b) 10 N. The sliding speed is 50 mm s^{-1}, and sliding distance is 20 m.

Application of the Hamrock and Dowson model to the nanotribology results would imply that lubricant films are 10^{-3}–10^{-5} nm thick at the nanoscale, which is of course physically unreasonable and reflects the inappropriateness of the model for a single asperity. It is known that the Hamrock and Dowson model becomes unreliable if the film thickness is very small, the solid surfaces are very smooth, or at low speeds.[54] The nanotribology tests satisfy all these conditions. Nonetheless, the tiny film thicknesses, combined with the nature of the nanotribology data strongly imply that that system is in the boundary regime.

At low loads (2 N) the macroscale friction coefficient is highest for pure hexadecane (0.11), and decreases when IL is added. The friction coefficient for 0.01 mol% IL is 0.06, the same as the pure IL, *cf.* Table 2. Increasing the IL concentration in hexadecane at this load does not further reduce this (low) friction coefficient. In the mixed regime that operates here, the added IL could reduce friction either by increasing the viscosity, which will keep the surfaces further apart during sliding, or by forming a boundary layer that prevents (intermittent) contact between surfaces, or both. It is not possible to discriminate between these two possibilities on the basis of friction coefficients alone. However, the zero shear viscosity of the IL–hexadecane mixtures shown in Table S1 in the ESI,† reveals that the viscosity increase between pure hexadecane and 1.0 mol% IL is negligible at least in the bulk.

Therefore, excluding a confinement induced viscosity increase, it is more likely that boundary layer formation is the more important effect.

When the normal load is increased to 10 N marginal changes in the friction coefficients of both the pure hexadecane and the pure IL are noted. At this load, the friction coefficients of the 0.01 mol% and 1.0 mol% IL–hexadecane mixtures are the same as pure hexadecane, *i.e.* the added IL has no measureable effect. This indicates that the boundary layers formed at these concentrations are too weak to prevent surface contact.

For 10 mol% IL at 10 N load, the friction coefficient is initially similar to that of the pure IL but at 6 m increases rapidly to a maximum of $\sim$0.17, and then gradually falls to the pure hexadecane value. The distance at which the rapid increase occurred varied between 4 m and 6 m for different experiments. This rapid change suggests that the boundary layer is ruptured suddenly and cannot reform. Most likely, natural fluctuations in the boundary layer integrity permit ball surface contact at some point, resulting in contact events that create debris causing the friction to increase. This point is expanded upon below. Similar boundary layer displacement was not observed for the 10 mol% IL–hexadecane mixture in the nanotribology tests. However, the total sliding distance in the nanotribology test is less than 1 mm, which is three orders of magnitude smaller than the macrotribology test distance.

Macroscale wear

Wear is the removal (or displacement) of material from a surface when subjected to contact and relative motion with another surface.[55] Wear is negligible during nanotribology tests as the sliding distance is typically very short ($<$1 mm), but it is often significant for macrotribology measurements. In this study, the wear tracks on the silica surfaces after the ball-on-disk experiments were studied by SEM, shown in Fig. 3. No detectable wear was observed for the pure IL at both 2 N and 10 N (not shown), consistent with excellent lubrication performance. This is likely because the lubrication by the pure IL is in the hydrodynamic regime; the thick lubricant film protects the surfaces well from wear.

At low normal load (2 N) the wear track is obvious for pure hexadecane, but becomes less pronounced with increasing IL concentration, *cf.* Fig. 3. The width of the wear track for pure hexadecane is $\sim$100 μm. This decreases slightly for the IL–hexadecane mixtures to 80–90 μm. Some wear debris was found within the wear track for pure hexadecane (Fig. 3a) but not for the IL–hexadecane mixtures (Fig. 3b–d).

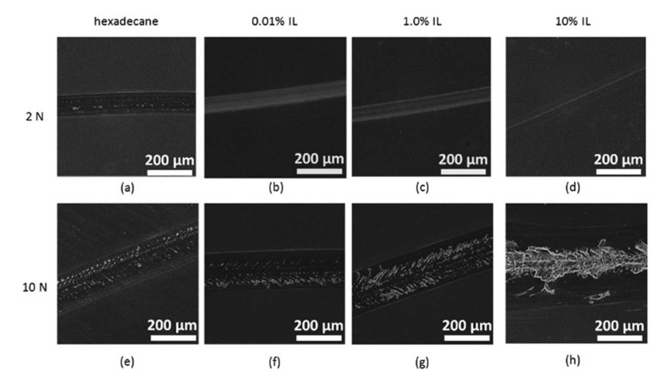

Fig. 3 SEM images of the contact area of silicon wafers lubricated with hexadecane (a and e), 0.01% IL (b and f), 1.0% IL (c and g) and 10% IL (d and h) at 2 N (a–d) and 10 N (e–h).

At 10 N significantly more debris is noted in the contact area and the widths of the wear tracks increase to $\sim$200 μm. At this high load the IL boundary layers can be penetrated, resulting in surface contact and thus ''debris''. Surprisingly, the debris and wear track width increase with IL concentration in the mixtures, and are most obvious for 10 mol%. This fact, combined with the characteristic, regular herringbone pattern suggests that the ''debris'' may be the result of a stick – slip phenomenon, and rather have the character of a tribofilm. The fact that the viscosity of the 10 mol% sample is almost double compared to lower concentration ones (ESI,† Table S3) may also lead to a reduction in the rate at which debris is naturally displaced from the wear track. With regard to Fig. 2b, friction coefficients at low IL concentrations are similar to that for hexadecane because the weakly formed IL boundary layers do not prevent surfaces from coming into contact. However, for 10 mol% IL, the presence of debris in the wear track leads to effective surface contact *via* ball – debris/tribofilm – substrate. Even if boundary layers are present on the ball and the substrate, any layer adsorbed to debris will be weak due to the debris' surface roughness disrupting the integrity of the film. Similar effects have been noted previously for pure IL–solid interfaces.[56] The absence of an effective boundary layer on the debris surface means that contact between the ball and the silicon wafer is relatively unimpeded. This leads to friction coefficients commensurate with those found at lower IL concentrations where the weak boundary layers do not prevent contact between the ball and substrate. Tribofilms of phosphorus compounds have been observed in several IL and IL–oil mixture macrotribology studies.[5,16,32–35] As both the cation and anion are based on the phosphonium ion, it is difficult to determine whether decomposition of the cation, anion or both is responsible for the film. In this work, significant formation of tribofilms would also locally deplete the IL concentration, and prejudice the ability of the boundary layer to heal *via* diffusive adsorption.

If present, water contamination can lead to wear in ILs.[54] The pure IL and oil used in this work are both highly hydrophobic, but due to its ionic nature the IL is slightly more hydrophilic, so the water content is expected to increase more strongly for higher IL concentrations. To rule this factor out, a 10 mol% IL in hexadecane sample was left open to the atmosphere for 30 minutes, which is more than five times longer than the time required for the ball on disk macrotribology experiments. Over this period the water content increased from 0.05 wt% to 0.25 wt%, which is still a low value. However, this increase, combined with the fact that no wear was detected for the pure IL, suggests that water contamination is not a key factor for wear in these systems, at least over the timescales probed. However, water contamination cannot categorically be ruled out as a contributor to wear, and future experiments will systematically examine the effect of water concentration on wear.

Fig. 4 shows the wear depth for dry contact, pure hexadecane and IL–hexadecane mixtures at normal loads of 2 N and 10 N.

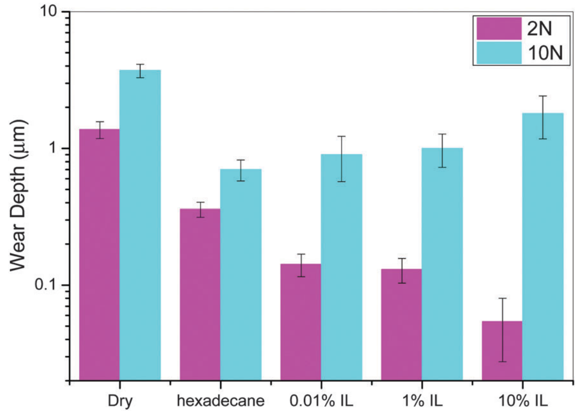

Fig. 4 Wear depth for the IL–hexadecane mixtures.

Data is not shown for the pure IL as no wear was apparent. The wear depth is greater at 10 N than at 2 N for all systems. Compared to dry contact, pure hexadecane decreases the wear depth by ~90% and ~80% at 2 N and 10 N, respectively. Added IL further decreases the wear depth at 2 N. For example, the wear depth for 10 mol% IL is only 1/3 of that for pure hexadecane. These results are consistent with the IL boundary layer remaining in place at low loads. However, when the normal load is increased to 10 N, the wear depths for the three IL–hexadecane mixtures are higher and increase with concentration. This is consistent with the morphology of the wear tracks (Fig. 3e–h) and the friction coefficient *vs.* distance data (Fig. 2b).

Conclusions

The nano- and macro-tribological performances of a series of IL–hexadecane mixtures at silicon surfaces have been examined under initially similar Hertzian contact stresses. At the nanoscale the system is in the boundary layer lubrication regime and lubricity is primarily a function of the integrity of the adsorbed layer. At low bulk IL concentrations ($\leq$1.0 mol%) the lower surface excess of ions in the boundary layer leads to a fragile boundary layer that is ruptured at high force. For higher IL concentrations ($\geq$2.0 mol%) the higher adsorption leads to a more robust boundary layer, and the lateral forces are similar to that of the pure IL.

For the macroscale ball-on-disc measurements the IL–hexadecane mixtures are in the mixed regime and the pure IL is in the hydrodynamic regime. The pure IL is an excellent lubricant at all loads. However, the performance of the IL–hexadecane mixtures is load dependant and the boundary layer robustness is the key determining parameter as it was for the nanoscale. As far as we are aware, this is the first time that nanotribological measurements have been combined with macroscopic friction studies to unambiguously demonstrate that the boundary layer behaviour is responsible for the macroscale friction performance for IL lubricants. At low loads, all of the IL–hexadecane mixtures tested performed as well as the pure IL, and significantly better than pure hexadecane, due to IL boundary layers adsorbing to the sliding surfaces. At high loads, the boundary layer for IL–hexadecane mixtures was immediately ruptured leading to surface contact, friction and formation of tribofilms and debris, for all but the 10 mol% mixture; the 10 mol% solution performed as well as the pure IL for sliding distances extending to between 4 m and 6 m until it too failed. Future work will examine how to achieve higher effective IL concentrations in oil and whether this is sufficient to prevent surface contact over longer distances, and investigate the nature of the tribofilms formed.

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